

**Characterization of the Building Materials in the De Lesseps Statue Base
and the Church of Saint Eugenie, Port Said, Egypt.**

توصيف مواد البناء في قاعدة تمثال ديليسبس وكنيسة القديسة أوجيني ببورسعيد، مصر.

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Abstract:

The city of Port Said, Egypt, contains many historic buildings that are at risk of deterioration due to natural and human factors. Conservation and preservation plans for these facilities require analytical studies of the building materials used to reach the most appropriate techniques and materials that will be applied in restoration and conservation work. This research will shed light on the chemical and mineral description and the precise composition of the building materials used in the construction of the de Lesseps statue base and the Church of Saint Eugenie in Port Said, which dated back to the nineteenth century AD. The examination and analysis work was based on four techniques: USB digital microscopy, Scanning Electronic Microscope (SEM) to determine the precise structure and surface morphology of materials with the use of the EDX unit to determine their constituent elements, X-ray fluorescence (XRF) analysis to determine the chemical elements of materials, and Raman- Spectroscopy analysis to determine the molecular composition of materials. It was concluded that limestone is main material, which used in construction of the de Lesseps statue base, two types were used: micrite limestone and biosparites limestone with the use of a hydraulic lime binding mortar. On the other hand, sandy limestone was used together with red bricks (bricks) in the construction of the Church of Saint-Eugenie, with the use of a cement and sand mortar (marine sand). Lime, gypsum and sand mortar was also used in the cladding of the western facade of the church.

Keywords: Port Said, De Lesseps statue base, Saint Eugenie Church, SEM-EDX, XRF, MRS.

الملخص:

تحتوي مدينة بورسعيد في مصر على العديد من المباني التاريخية المهددة بالتدهور بسبب العوامل الطبيعية والبشرية. تتطلب خطط الحفاظ على هذه المرافق وصيانة أثارها دراسات تحليلية لمواد البناء المستخدمة للوصول إلى أنسب التقنيات والمواد التي سيتم تطبيقها في أعمال الترميم والصيانة. سيليقي هذا البحث الضوء على الوصف الكيميائي والمعدني والتركيب الدقيق لمواد البناء المستخدمة في بناء قاعدة تمثال ديليسبس وكنيسة القديسة أوجيني في بورسعيد والتي يعود تاريخها إلى القرن التاسع عشر الميلادي. استندت أعمال الفحص والتحليل على أربع تقنيات: المجهر الرقمي USB، والمجهر الإلكتروني الماسح (SEM) لتحديد التركيب الدقيق ومورفولوجيا سطح المواد مع استخدام وحدة EDX لتحديد العناصر المكونة لها، وتحليل فلورسنت الأشعة السينية (XRF) لتحديد العناصر الكيميائية للمواد، وتحليل مطيافية رامان لتحديد التركيب الجزيئي للمواد. خلصت الدراسة إلى أن الحجر الجيري هو المادة الرئيسية المستخدمة في بناء قاعدة تمثال دي ليسبس، وقد استخدم نوعان: الحجر الجيري الميكريتي والحجر الجيري البيوسباريتي، مع استخدام ملاط ربط هيدروليكي من الجير. من جهة أخرى، استخدم الحجر الجيري الرملي مع الطوب الأحمر في بناء كنيسة القديسة أوجيني، باستخدام ملاط أسمنتي ورملي (رمل بحري). كما استخدم الجير والجبس وملاط الرمل في تغطية الواجهة الغربية للكنيسة.

الكلمات المفتاحية: بورسعيد، قاعدة تمثال ديليسبس، كنيسة سانت أوجيني، SEM-EDX، XRF، MRS.

1. Introduction

Ancient buildings and structures represent a vital component of the tangible cultural heritage of nations and societies, preserving their identity and belonging¹. Conservation and maintenance efforts for these structures and buildings are a crucial step in preserving the memory of human civilization². These efforts require comprehensive knowledge of all aspects related to heritage through diagnostic and analytical studies to develop appropriate conservation plans³. Restoration work without diagnostic analysis often leads to unsatisfactory results and may even damage cultural heritage. Previous studies confirm that differences in the physical and chemical properties of original and conservation materials-such as mechanical strength, permeability, and durability-can cause compatibility issues^{4 5 6}.

Port Said is one of the Egyptian cities that contain a number of historic buildings and structures⁷. The city's establishment is linked to an important event in Egyptian history, the beginning of the excavation of the Suez Canal in 1859⁸. It is located in the northeastern part of the Arab Republic of Egypt and serves as the gateway to the Suez Canal on the Mediterranean Sea^{9,10}. Among the most significant historic landmarks in the city are the statue base of de Lesseps and the St. Eugénie Church, both dating back to the late 19th century, and are registered as archaeological sites under the Egyptian Antiquities Protection Law No. 113 of 1983 and its amendments. The base of the De Lesseps statue was registered as an antiquity by Ministerial Decree No. 228 of 1997, published in the Egyptian Gazette, Issue No. 23 of 1998. The Church of Saint Eugenie was registered as an antiquity by Ministerial Decree No. 561 of 2016, published in the

1 Abdulnasser Al-Zahrani, *Cultural Heritage: Its Nature, Threats, and Preservation*. 2017, Kingdom of Saudi Arabia: King Faisal Center for Research and Islamic Studies.

2 Isis Fahd, "Restoration and Preservation of Heritage in Italy: The Case Study of Orvieto and its Application in Palestine: The Case Study of Iraq Burin." 2010, An-Najah National University in Nablus-Palestine.

3 Cheitih Ezzeddine, "Reuse of Historical Landmarks: Building Yesterday's Structures for Today's Functions." 2011, Mentouri University Constantine-Algeria.

4 Moropoulou, A., et al., Strength development and lime reaction in mortars for repairing historic masonries. *Cement and Concrete Composites*, 2005, 27(2), 289-294.

5 Maravelaki-Kalaitzaki, P., et al., Hydraulic lime mortars for the restoration of historic masonry in Crete. *Cement and Concrete Research*, 35(8), 2005, 1577-1586.

6 Luque, A., et al., The use of lime mortars in restoration work on architectural heritage. *Materials, technologies and practice in historic heritage structures*, 2010, 197-207.

7 Ali, A.G.M., "Orientation and Vision for the Sustainability of Heritage Sites: A Case Study of Port Said City." *International Journal of Conservation Science*, 2021. 12(4): pp. 1343-1354.

8 Badr Abdelaziz, "Architectural Styles of Port Said City during the Muhammad Ali Pasha Family Era." *The Arab Archaeologists Association Journal*, 2011. 14(14): pp. 666-707.

9 Warda Hassan, "Development in Port Said Governorate: A Geographical Study." 2010, Al-Azhar University: Faculty of Humanities - Al-Azhar University - Egypt.

10 General Organization for Urban Planning, "Environmental Perspective for the Urban Development Strategy at the National Level: Suez Canal Region." 2014, Ministry of Housing, Utilities, and Urban Communities: Cairo-Egypt.

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Egyptian Gazette, Issue No. 121 of 2017. Limestone is the main building material for the base of the statue, while it was used along with fired bricks in the construction of the church.

This research deals with a diagnostic study of the building materials used in the construction of the De Lesseps statue base and the Church of Saint Eugenie, through several steps starting with visual examination on site, and then determining the chemical and mineral composition through the application of advanced methods such as scanning electron microscopy, X-ray fluorescence, and Raman spectroscopy, in order to understand the mechanisms of damage and reach conservation materials compatible and appropriate for the original materials^{11 12}.

1.1 De Lesseps statue base

In 1897, the Universal Company of the Suez Canal decided to erect a memorial to commemorate Ferdinand de Lesseps, the concessionaire of the canal's excavation, by constructing a statue overlooking the northern entrance of the canal¹³. The base was completed, and the statue was placed on November 17, 1899, coinciding with the thirtieth anniversary of the Suez Canal's opening to global navigation in 1869¹⁴. The base of the de Lesseps statue consists of two main parts: the statue's base and the forward pier, known as the promenade (see Figure 1A).

The base is located at the end of the northern promenade and is constructed entirely of limestone. It starts from the bottom with a circular base, 17 meters in diameter, and has a total height of 9.40 meters. The external frame is surrounded at the bottom by sixteen round stone pillars with rounded tops resting on concave curved bases. These bases merge at the bottom with the pillar cylinders and are connected by an iron chain that acts as a barrier to prevent falls. Access to the base was via five steps on the southern side of the base at the end of the promenade before the promenade's surface was elevated at the end of 1923 (see Figure 1B).

The pier is an extension of Palestine Street (François Joseph Street) within the Eastern District. Initially, it was at the same level as the street and currently occupies the city's northern corner. It starts from the south at the intersection of the city's two prominent streets, Palestine Street (François Joseph Street) and 23 July Street (Kitchener), and ends in the north, overlooking the fishing port basins in Port Said. It is a long pier extending from south to north, approximately 350 meters long at its end and about 316 meters long at the beginning of the statue's base, with a width of 6.30 meters

11 Binda, L., et al., Investigation procedures for the diagnosis of historic masonries. *Construction and Building materials*, 2000, 14(4), 199-233.

12 English Heritage Publishing 2006, *Identifying and Sourcing Stone for Historic Building Repair*.

13 Archives nationales du monde du Travail, Roubaix, Fonds de la Compagnie universelle du canal maritime de Suez(FRANMT_1995_060_89_03_0002). Available from: <https://recherche-anmt.culture.gouv.fr/ark:/1128833/60879>

14 Demetrios Khaldoubis, *Events and Memories of Port Said*. 1939, Alexandria-Egypt

and a height of 2.90 meters from the level of Palestine Street (François Joseph Street) at the time of its construction (see Figure 1C).

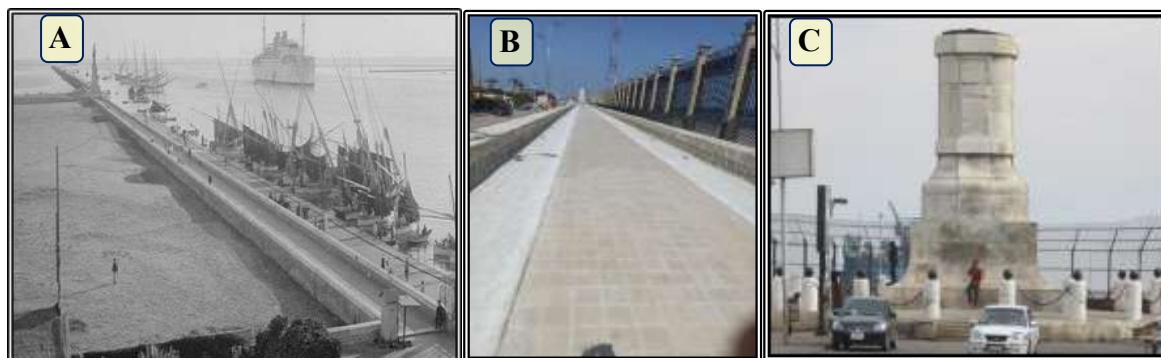


Figure 1: (a) shows the base of the de Lesseps statue and the promenade, taken from the Archives¹⁵; (b) shows the base of the statue; (c) shows the promenade with the base at the end (photographed by the researcher)

In 1923, the company carried out development works on the promenade, raising it by 75 cm above the street level, followed by an increase in the promenade's size¹⁶. Limestone was used in the construction of the promenade, noticeable in the limestone blocks (rubble) used in building the promenade's walls (parapets), and the limestone blocks used in finishing the surfaces of the walls.

1.2 Church of Saint Eugenie

Saint Eugénie Church was first inaugurated in 1868 to serve the Catholic communities in Port Said¹⁷. It was rebuilt with stone and brick in 1890¹⁸. The church is located at the intersection of Ahmed Maher Street (formerly Alexander the Great Street) and Ramses Street, Manshiyat Square (formerly De Lesseps Square) in the Eastern District.

The church follows the "basilica plan" influenced by European architecture¹⁹, characterized by its elongation and large area. It measures 40.75 meters in length from north to south, 22.40 meters in width from east to west, with a total area of 912.8 square meters, and is elevated by one meter above street level. It consists of one floor above the basement or (crypt) (see Figure 2A). The horizontal layout features symmetry and centrality, forming a rectangle extending from north to south. The eastern and western sides are longer than the northern and southern sides, divided into three wings by a series of columns perpendicular to the altar area. The central wing is the widest and

15 Archives nationales du monde du Travail, Roubaix, Fonds de la Compagnie universelle du canal maritime de Suez(FRANMT_1995_060_90_44_0003)

16 Minutes from the Port Said Municipal Council meeting on July 7, 1923.

17 Administration of Saint Eugénie Church in Port Said, Notebook on the Franciscans in Port Said: Port Said-Egypt.

18 National Archives, Letter from the Prime Minister's Office to the Governor General of the Canal, Record 125. 1890, National Archives: Cairo-Egypt.

19 Ezzat Qadous and Mohamed Abdel-Fattah, Coptic and Byzantine Antiquities. 2002, Alexandria-Egypt: Al-Hadary Press

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highest, aligned along a longitudinal axis that starts at the entrance in the north and ends at the altar in the south (see Figure 2B).

Saint Eugénie Church is rich in heritage values, represented in its architectural elements and diverse artistic possessions. It features Tuscan columns, prominent cornices extending horizontally along the facades, triangular pediments, architectural entablatures, panels, and various decorative and artistic elements such as shields, floral arches, and other decorative features adorning the external facades. The windows and doors have a rectangular shape with semi-circular arches, and circular windows appear above the church entrance. The church features various roofing styles: the northern and western wings have flat concrete roofs, while the central wing has a continuous longitudinal barrel vault topped by another roof in the form of a gable.

The church contains a tower located at the far south of the northwestern facade, adjacent to it, with an approximate height of 20 meters. The tower consists of a square base from the church floor level up to the roof, followed by a second floor slightly recessed carrying a rectangular balcony, then a third floor topped by a conical peak ending with a Latin cross on the outside. The nave of the church is a rectangular space of one floor, tiled with marble slabs, extending from north to south with regular dimensions, measuring 36.5 meters in length and 20.4 meters in width. It is divided into three wings, the widest being the central wing, extending 36.5 meters in length and 9.4 meters in width, divided by two colonnades, each containing seven semi-circular arches, perpendicular to the altar of the church from north to south, supported by eight square stone columns (see Figures 2C, 2D).

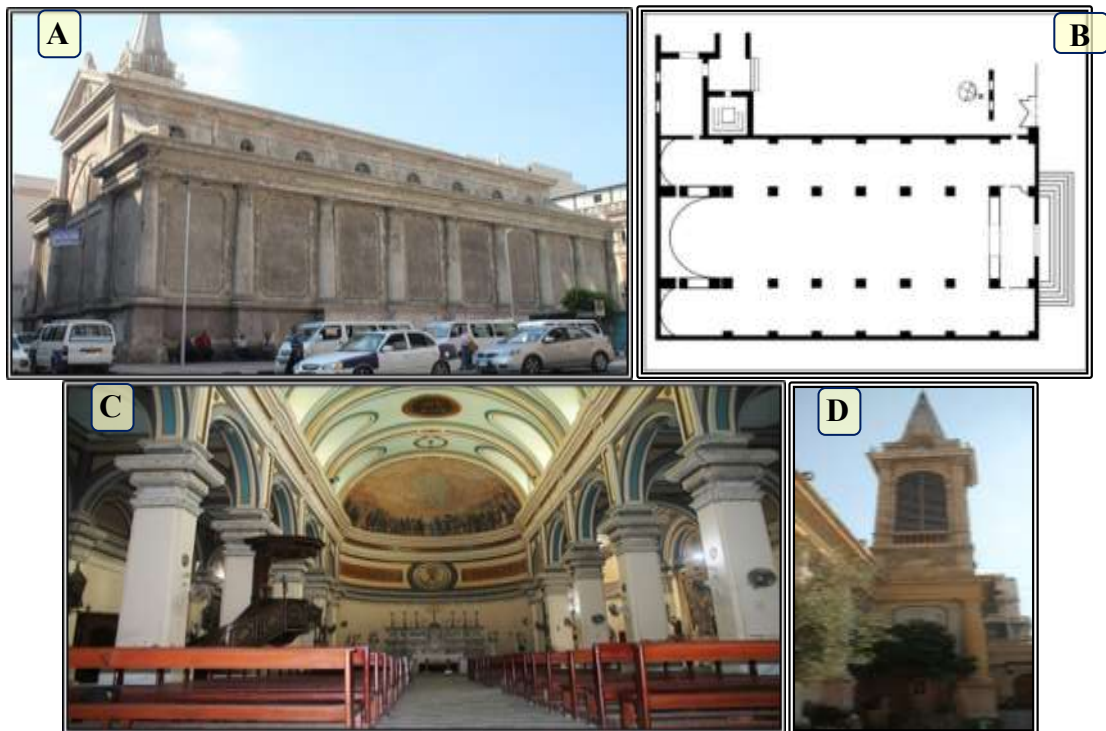


Figure 2: (a) Saint Eugénie Church, (b) shows the horizontal layout of the church (researched by the author); (c) shows the interior of the church; (d) shows the church tower (photographed by the author).

2. Materials and Methods

2.1 Materials

In this research, eight different samples were taken from various locations representing the building materials used in both the base of the de Lesseps statue and the Saint Eugénie Church. Samples A, B, C, and H represent materials from the statue base, while samples D, E, F, and G are from the church, as shown in Table 1 and Figure 3A, B.

Table 1 shows the type and location of the samples under study.

No	Sample Code	Sample Type	Sample Location
1	A	base stone	Southern façade of the base
2	B	Base bonding mortar	Eastern façade of the base
3	C	Promenade stone	Western wall of the base
4	D	Church building stone	Eastern wall of the church
5	E	Church building brick	Eastern wall of the church
6	F	Church building mortar	Eastern wall of the church
7	G	Church wall plaster	Western wall of the church
8	H	Part of the base stone veining	Northern facade of the base

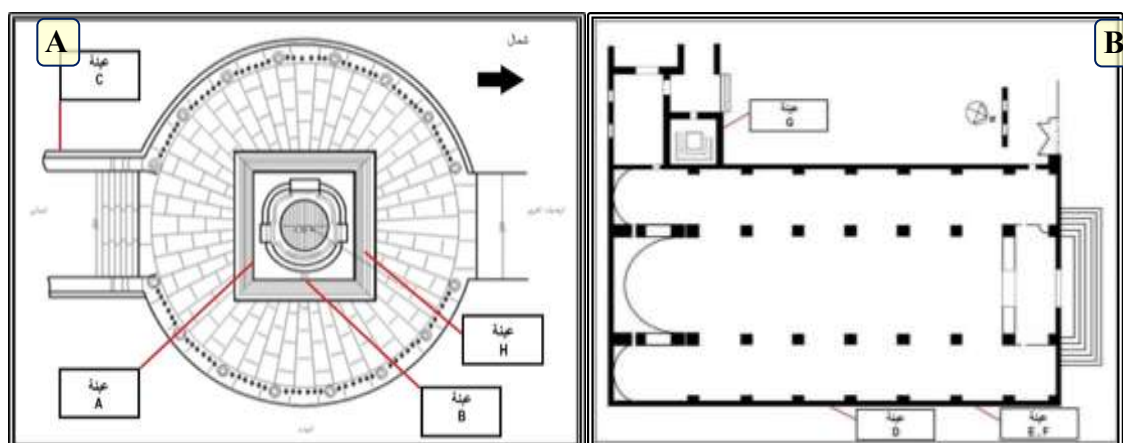


Figure 3: (a) Shows the locations of the base samples, (b) Shows the locations of the church samples (designed by the researcher).

2.2 Method

2.2.1 USB Microscopy

The portable digital microscope (USB Microscopy USB ROHS, M8704-1000, China) with magnifications of 100x and 1000x was used to study the surface morphology of the samples. The digital microscope is a simple and cost-effective method for obtaining high-magnification color microscopic Figures.

2.2.2 Scanning Electronic Microscope (EDX)

The Scanning Electron Microscope (SEM) (JEOL JSM-6510 LV, JEOL Ltd., Japan, ACCEL_20KV, MAG 4300, SIGNAL SEI, WD 10 mm) was used at Mansoura

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University to examine the fine structure, with the use of an EDX unit to identify chemical elements^{20 21}, the sample coated with gold then examination by SEM.

2.2.3 X-ray fluorescence (XRF)

The samples were analyzed using X-Ray Fluorescence (XRF) Spectroscopy (Handheld XRF model S1 TITAN from Bruker, using a 50 kV power supply and operated at 40 μ Amps) at Damietta University to detect the elemental composition of the samples²².

2.2.4 Raman Spectroscopy

The analysis was also conducted using a Confocal Raman microscope (Jasco NRS-4500, Tokyo, Japan) at Damietta University to identify the molecular and mineral composition of the samples²³, For both Raman data acquisition and processing, Jasco spectroscopy suite software was used Micro-Raman spectroscopy has several advantages to make it the first choice for combined, analytical procedures and a particular advantage quantitative studies is the linear relationship between species concentration and band intensity^{24 25}.

3. Results

3.1 Examination and Analysis of the Building Materials of the de Lesseps Statue Base

3.1.1 Imaging and Examination with Digital USB Microscopy

The stones of the base appear to be limestone, characterized by the homogeneity and cohesion of their grains, forming a single, polished, and solid mass. Figures captured with the USB Digital Microscope (USB D.M) for sample A show the grains'

20 İpekog˘lu, B., et al., Assessment of material use in relation to climate in historical buildings. Building and environment, 2007. 42(2): p. 970-978.

20 Batool, F., et al. Material characterization of a historical wall by using destructive and non-destructive techniques. in Structures. 2022. Elsevier.

21 Mabrouk, Nabil, and Yosr Elsayed. "Archaeometrical Study of a Rare Embroidered and Appliqued Leather Tapestry from the Safavid Artworks. Part Ii: Colored Leather." Mediterranean Archaeology and Archaeometry 20.3 (2020): 1-1.

22 Batool, F., et al. Material characterization of a historical wall by using destructive and non-destructive techniques. in Structures. 2022. Elsevier.

23 de Ferri, L., et al., A multi-technique characterization study of building materials from the Exedra of S. Nicolò l'Arena in Catania (Italy). Journal of Building Engineering, 2019. 23: p. 377-387.

24 Schulte, Franziska, et al. "Raman spectroscopy of synthetic organic pigments used in 20th century works of art." Journal of Raman Spectroscopy: An International Journal for Original Work in all Aspects of Raman Spectroscopy, Including Higher Order Processes, and also Brillouin and Rayleigh Scattering 39.10 (2008): 1455-1463.

25 Elsayed, Yosr. "A Multi analytical Study of a 20th Century Oil Painting of King Farouk at The Egyptian Agricultural Museum. Journal of the faculty of archaeology: Cairo university, 18(28),2025, 489-465 .

cohesion, with some veining that may be calcite, and signs of weathering and surface irregularities on the sample exposed to the atmosphere (Figure 4A).

On the other hand, the stones used in constructing the pier are composed of conglomerates of marine sediments (shells and skeletons), with sand grains bonded together in a single structure. This indicates that the stones are organic limestones. Magnified Figures with the USB Digital Microscope (USB D.M.) clearly show the sand grains and organic remains with the binding material of calcite (Figure 4B).



Figure 4: USB Microscope: (a) for sample A, base stone, (b) for sample C, pier stone, (c) for sample B, bonding mortar.

The bonding mortar, as observed through the digital microscope, appears to consist of a binding material that could be cement or hydraulic lime due to its dark gray color. The phases of dicalcium silicate (belite or larnite) (C₂S) exhibit a yellow or beige color, while tricalcium silicate (alite) (C₃S) shows a dark gray color. These are characteristic phases of hydraulic lime and cement ²⁶(Figure 4C).

3.1.2 SEM (EDX) Analysis

The results of the analysis using the EDX unit, according to Table 2, show that the base stone (sample A) is primarily composed of the elements oxygen and calcium (Ca, O), confirming that the stone used in the base construction is high-calcium limestone. Sodium and chlorine were detected as secondary elements, indicating the presence of halite salt (Halite) due to surface deposition from weathering, which inevitably leads to the weakening and deterioration of the building stone. Silicon was present as an impurity due to the stone's formation conditions. The results for sample H, representing the crystal veining in the base stone, revealed calcite crystals (CaCO₃), with copper, zinc, and aluminum present as impurities.

On the other hand, the analysis results for the pier stone (sample C) indicated that oxygen and calcium are the main elements composing the stone, confirming that the pier stone is high-calcium limestone. Silicon appeared as a secondary element, and magnesium was detected, possibly indicating the presence of the mineral dolomite

26 Jebbawy, M., et al., First historical cements in France from Boulogne sur Mer: Investigations on 19th century's manufactured cement-based monuments. *Journal of Microscopy*, 2022. 286(2): p. 63-68.

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(CaMg(CO₃)₂) as a component of the stone. Chlorine suggested the presence of halite salt, while iron appeared as an impurity.

As for sample B, representing the bonding mortar used in construction, the main elements detected were oxygen, calcium, and silicon, with aluminum, iron, magnesium, and potassium present as secondary elements. This suggests that the mortar was hydraulic lime or cement rather than ordinary lime²⁷ (Figure 5A, B, C, D).

Table 2 shows the EDX analysis results for samples A, B, C, and H:

Sample	O (%)	Ca (%)	Si (%)	Mg (%)	Fe (%)	Al (%)	Cl (%)	Na (%)	K (%)	C (%)	Zn (%)	Cu (%)
A	62.92	33.37	0.57	-	-	-	1.03	2.11	-	-	-	-
B	54.47	29.37	9.10	0.75	1.36	2.40	2.21	-	0.34	-	-	-
C	53.4	36.38	5.67	2.32	1.16	-	1.07	-	-	-	-	-
H	43.54	36.14	-	-	-	0.06	-	-	-	19.3	0.33	0.62

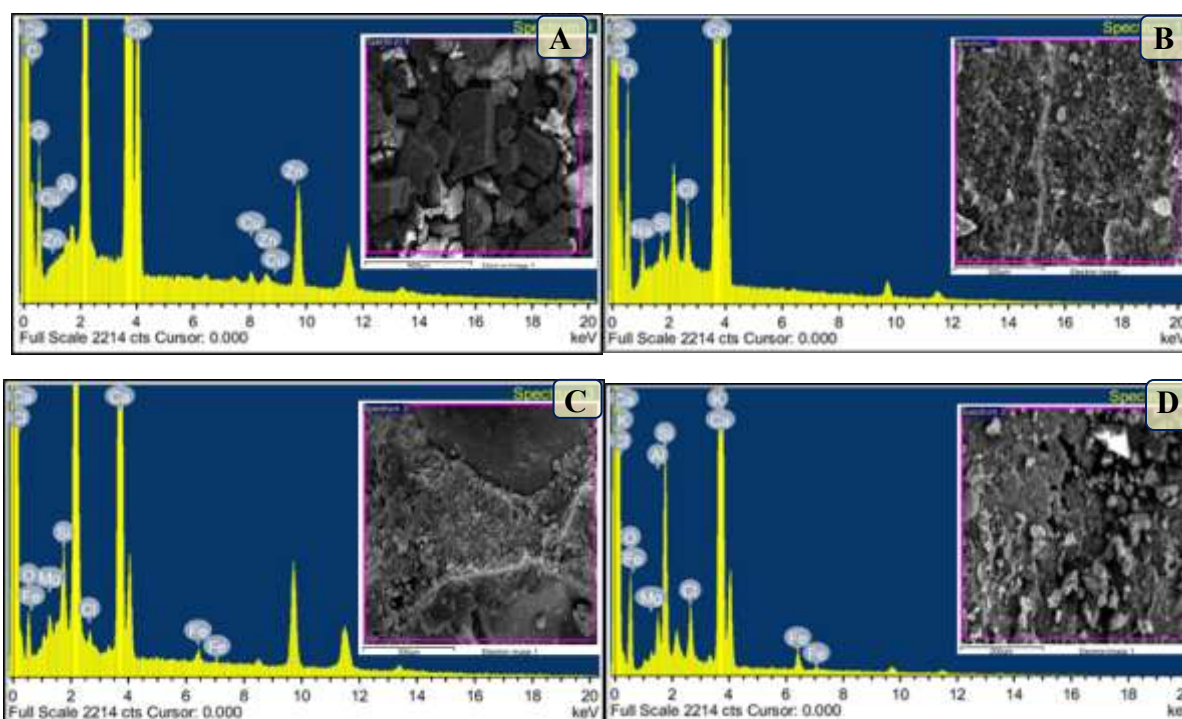


Figure 5: SEM-EDX Analysis Pattern: (a) for sample A (base stone), (b) for sample H (veins in the base stone), (c) for sample C (pier stone), (d) for sample B (bonding mortar).

The results of the examination and imaging using the Scanning Electron Microscope (SEM) showed the homogeneity of the internal structure of the base stone sample. The micrite network (micrite CaCO₃) forms the primary internal structure of the stone, as shown by the number (1 in Figure 6A). The formation of compact, non-

²⁷ Figueiredo, et al., Chemical and physical characterisation of three NHL 2 binders and the relationship with the mortar properties. in REHABEND 2016 Euro-American Congress: Construction Pathology, Rehabilitation Technology And Heritage Management (6th REHABEND Congress). 2016.

surfaced micrite units, from polyhedral particles to completely fused anhedral particles, with distinct contacts between the crystals but with little porosity, is observed. This fine texture is known as a "non-porous crystalline mosaic texture"²⁸. Surface traces of halite salt (NaCl) crystals are also seen in scattered locations, as indicated by number 2, suggesting that the stone's deterioration mechanism due to salts is active under suitable conditions. The presence of fine pores, attributed to weathering and decomposition factors, is indicated by the arrows in Figure 6B.

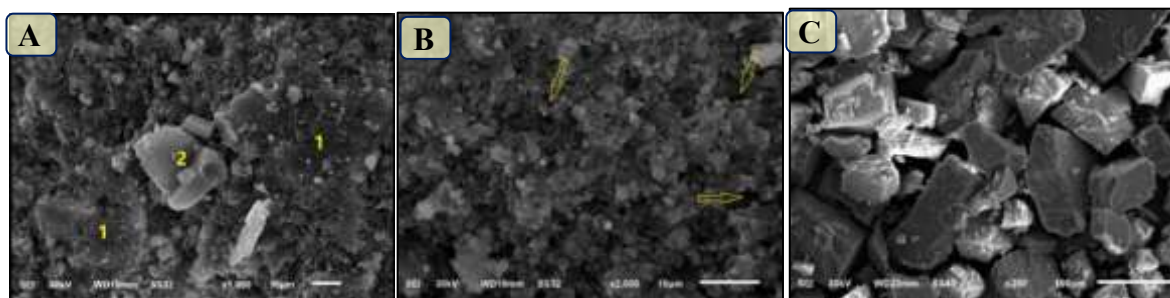


Figure 6: SEM Examination: (a) for sample A (base stone) where number 1 indicates the micrite network and number 2 indicates halite crystals, (b) for sample A (base stone) where the arrows indicate the pores due to weathering, (c) for sample H where the veins consist of calcite crystals.

The calcite crystals, the main component of the crystal veins in the base stone, are clearly visible (Figure 6C). This confirms that the stones used in constructing the base are re-crystallized micrite, calcite limestone, which originates from carbonate mud forming a dense homogeneous micrite with clumped calcite crystals forming veins^{29 30}.

The SEM examination reveals that the fine internal structure of the pier stone sample consists of quartz grains and marine shell and skeleton remains (Figure 7A), bonded by calcite deposition (CaCO_3). Figure 7B shows the quartz grains (indicated by number 1) and the calcite deposits acting as a binding material (indicated by number 2). Another form of calcium carbonate observed is the elongated aragonite crystals (indicated by number 3). Halite salt crystals (NaCl) are also visible on the surface and within the pores (indicated by number 4).

28 de Periere, M.D., et al., Morphometry of micrite particles in cretaceous microporous limestones of the Middle East: Influence on reservoir properties. *Marine and Petroleum Geology*, 2011. 28(9): p. 1727-1750.

29 Oates, J.A., *Lime and limestone: chemistry and technology, production and uses*. 2008: John Wiley & Sons.

30 Adefris, D., et al., Petrography and stable oxygen and carbon isotopic composition of the Antalo Limestone, Mekelle Basin, Northern Ethiopia: implications for marine environment and deep-burial diagenesis. *Carbonates and Evaporites*, 2020. 35: p. 1-17.

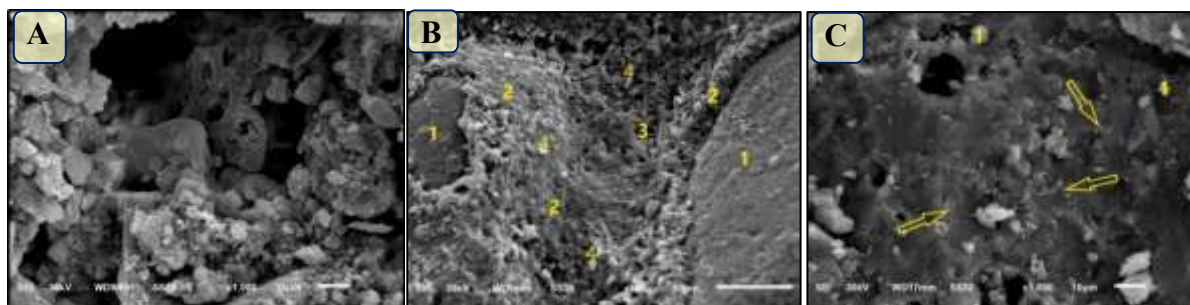


Figure 7: SEM Examination: (a) for sample C (pier stone), showing marine shell remains, (b) for sample C (pier stone), where number 1 indicates quartz, 2 indicates calcite deposits, 3 indicates aragonite, and 4 indicates halite, (c) for sample B (bonding mortar), with arrows indicating calcium silicate hydrate and number 1 indicating halite.

This confirms that the stone used in the construction of the promenade (pier) is organic limestone (Biosparites limestone), composed of organic remains (Bioclasts) and quartz sand, bonded together by a network of calcite deposits^{31 32}, as shown in the Figures.

The SEM analysis of the bonding mortar (sample B) revealed that it consists of an interconnected structure of calcium and silicon. The surface of the sample shows calcium silicate hydrate (CSH) in the form of fine fibers, indicated by the arrows in (Figure 7C). The presence of halite salt (NaCl) as deposited crystals, indicated by number 1, suggests one of the weathering agents causing deterioration. Large pores, possibly resulting from the weathering and decomposition of mineral compounds, are also visible.

3.1.3 X-ray fluorescence (XRF) Analysis

The elemental analysis of the base and promenade stone samples revealed high levels of calcium oxides, confirming that the stones used in construction are limestone, where calcium oxides form the primary component. Silicon oxides appeared as a secondary component in the stone samples but in varying proportions, demonstrating the difference in the type of limestone used for the base versus the promenade. The base stone is predominantly composed of calcium oxides in the form of calcite and carbonates (79.5% CaO with error margins $\pm 0.12\%$ - 6.2% $\text{SiO}_2 \pm 0.38\%$), resulting in strong and durable stone with less porosity. In contrast, the promenade stone shows a higher proportion of silicon oxides, indicating the presence of sand with calcium oxides forming the binding material and organic remains (53.7% CaO $\pm 0.09\%$ - 26.5% $\text{SiO}_2 \pm 0.33\%$), resulting in a coarser stone texture with increased porosity (Figure 8A, B, C), table 4.

31 Oates, J.A., Lime and limestone: chemistry and technology, production and uses.

32 Adefris, D., et al., Petrography and stable oxygen and carbon isotopic composition of the Antalo Limestone, Mekelle Basin, Northern Ethiopia: implications for marine environment and deep-burial diagenesis. Carbonates and Evaporites.

Magnesium oxides in the base and promenade stone samples confirm that they are high-calcium limestones. Limestone with more than 20% by weight of dolomite ($\text{CaMg}(\text{CO}_3)_2$) is classified as dolomitic lime, while limestone with less than 5% MgCO_3 is classified as high-calcium lime^{33 34}.

The presence of other elements as impurities, such as iron oxides, was noted. The presence of sulfur elements is attributed to the chemical weathering of limestone. The presence of titanium oxides as impurities suggests enhanced resistance to future biological degradation, acting as a biocide³⁵.

The presence of chlorine indicates the presence of halite salts, with the base and promenade's proximity to the sea facilitating constant salt deposition on the stones, accelerating deterioration. Strontium as an impurity in the promenade stone indicates the formation of marine limestone, with a concentration of approximately 8 milligrams per liter in Mediterranean seawater³⁶. Groundwater connected to the sea may also contain strontium.

The XRF analysis of the mortar revealed a high percentage of calcium oxides ($70.7\% \pm 0.07\%$) with the presence of magnesium oxides ($10.8\% \pm 3.1\%$), indicating the use of limestone with a high dolomite content in the mortar's production. The presence of silicon, aluminum, and iron oxides confirms that the mortar used was hydraulic lime or cement, as these oxides constitute the mineral composition of the clay added to lime to produce hydraulic lime or cement. The Hydraulic Index (H.I.) and Cement Index (C.I.) were calculated according to the equations in Table (3). C.I. values less than 0.3 indicate aerial (ordinary) lime, values between 0.3 and 0.5 indicate low hydraulic lime, values between 0.5 and 0.7 indicate moderate hydraulic lime, values between 0.7 and 1.1 indicate high hydraulic lime, and values greater than 1.1 indicate cement^{37 38 39}. The C.I. value of 0.405 suggests that the mortar used was not ordinary aerial lime.

33 Thirumalini, S., et al., Knowing from the past–Ingredients and technology of ancient mortar used in Vadakumnathan temple, Tirussur, Kerala, India. *Journal of Building Engineering*, 2015. 4: p. 101-112.

34 Batool, F., et al. Material characterization of a historical wall by using destructive and non-destructive techniques.

35 Morillas, H., et al., Composition and porosity study of original and restoration materials included in a coastal historical construction. *Construction and building materials*, 2018. 178: p. 384-392.

36 Bernat, M., et al., Barium and strontium concentrations in Pacific and Mediterranean Sea water profiles by direct isotope dilution mass spectrometry. *Earth and planetary science letters*, 1972. 16(1): p. 75-80.

37 Thirumalini, S., et al., Knowing from the past–Ingredients and technology of ancient mortar used in Vadakumnathan temple, Tirussur, Kerala, India.

38 Callebaut, K., et al., Nineteenth century hydraulic restoration mortars in the Saint Michael's Church (Leuven, Belgium): Natural hydraulic lime or cement? *Cement and Concrete Research*, 2001. 31(3): p. 397-403.

39 Figueiredo, C., et al., Chemical and physical characterisation of three NHL 2 binders and the relationship with the mortar properties.

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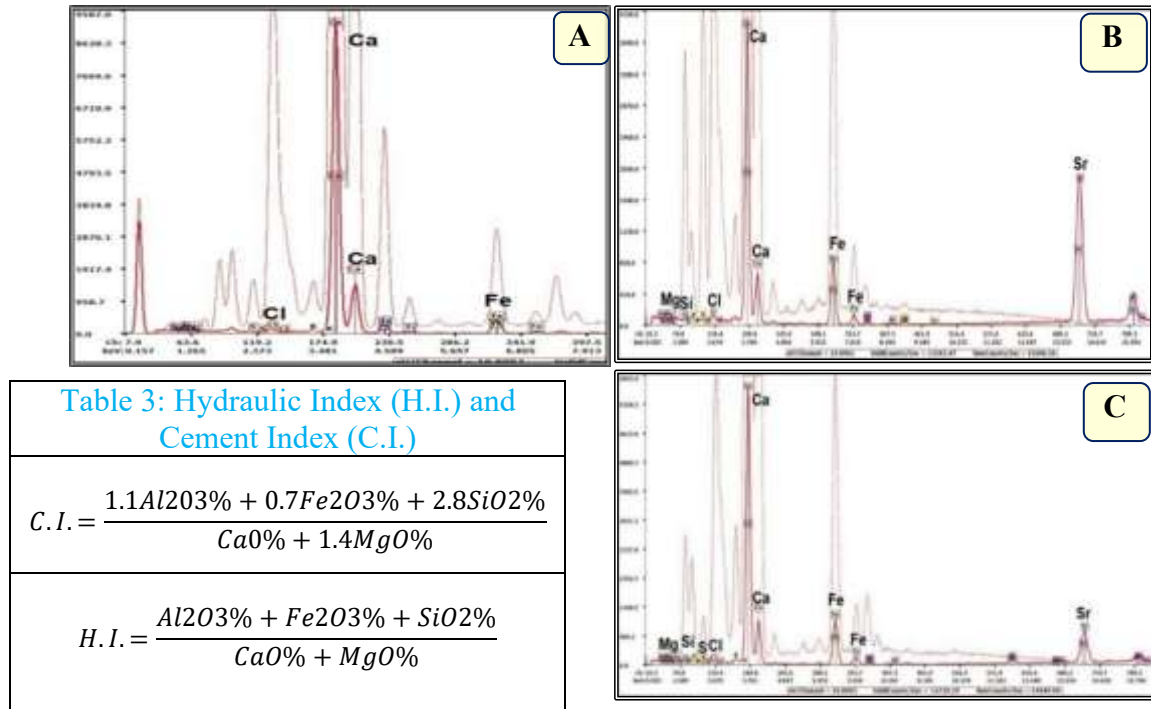


Figure 8: XRF Pattern Results for: (a) Sample A (base stone), (b) Sample C (pier stone), (c) Sample B (bonding mortar).

Table 4 shows the XRF analysis results for samples A, B, and C:

Comp.	SiO ₂	CaO	Al ₂ O ₃	MgO	Fe	P ₂ O ₅	S	Ti	Cl	others
A	6.2	79.5	3.9	1.97	0.6	1.2	0.31	2.4	3.7	0.22
B	10.7	70.7	3.4	10.8	1.4	0.85	0.18	0.14	0.85	0.98
C	26.5	53.7	2.9	4.6	1.4	0.51	4.7	0.17	4.3	1.22

3.1.4 Raman Spectroscopy Analysis

3.1.4.1 Building stones

The analysis results for sample A, the base stone, indicate that it is composed of limestone, with calcium carbonate (CaCO₃) as the primary component in the form of calcite. Dolomite and quartz minerals are also present in the stone, along with hydromagnesite, a natural component of limestone⁴⁰, and phosphate as an impurity. The analysis results for sample C also confirm it is limestone, with the presence of dolomite, quartz, anorthite, brucite, hematite, calcium phosphate, and ettringite.

⁴⁰ Hollingbery, L. and T.R. Hull, The fire retardant effects of huntite in natural mixtures with hydromagnesite. Polymer Degradation and Stability, 2012. 97(4): p. 504-512.

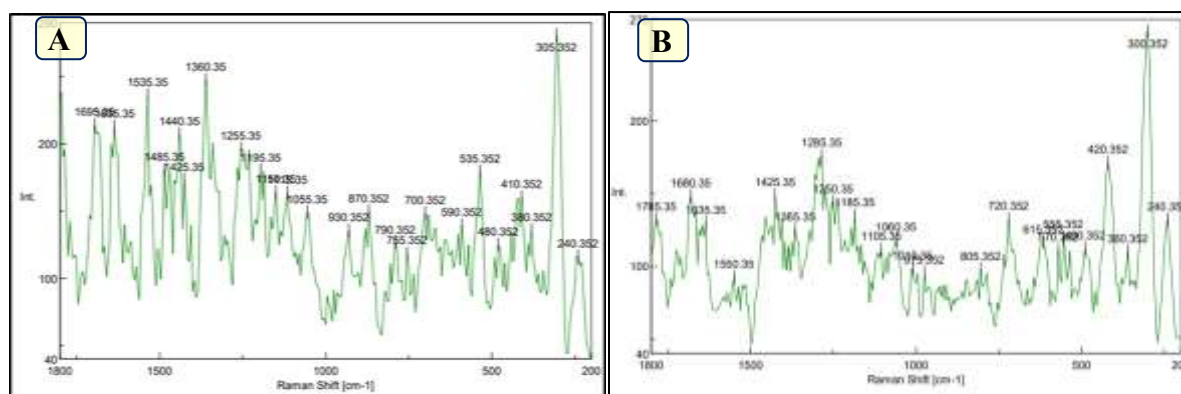


Figure 9: Raman Spectroscopy Results: (a) for sample A (base stone), (b) for sample C (pier stone).

It is worth noting that the presence of calcium sulfate, calcium oxalate, calcium nitrate, and sodium sulfate compounds indicates the chemical weathering of limestone⁴¹⁴². The presence of black carbon shows the impact of air pollution on the stone surfaces. Halite salt in the stone samples results from surface deposition (Figure 9A, B) (see Table 5).

Table 5: Raman Analysis Results for Building Stones (Samples A-C) and includes the chemical composition, characteristic wavelengths, and main references.

No.	Composition	Sample A	Sample C	Ref.
1	Calcite-Aragonite-Vaterite CaCO_3	700-755-1090-1635	705-720- 1090-1635	45 44 43
2	Dolomite $\text{CaMg}(\text{CO}_3)_2$	305-1440	300-1105	49 48 47 46

41 Allen, G., et al., Role of NO₂ and SO₂ in degradation of limestone. British Corrosion Journal, 2000. 35(1): p. 35-38.

42 Cezar, T.M., Calcium oxalate: a surface treatment for limestone. Journal of Conservation and Museum Studies, 1998. 4: p. 6-10.

43 Gunasekaran, S., et al., Raman and infrared spectra of carbonates of calcite structure. Journal of Raman Spectroscopy: An International Journal for Original Work in all Aspects of Raman Spectroscopy, Including Higher Order Processes, and also Brillouin and Rayleigh Scattering, 2006. 37(9): p. 892-899.

44 Topuz, G., et al., Partial high-pressure aragonitization of micritic limestones in an accretionary complex, Tavşanlı Zone, NW Turkey. Journal of Metamorphic Geology, 2006. 24(7): p. 603-613.

45 Garg, N., Raman spectroscopy for characterizing and determining the pozzolanic reactivity of fly ashes. 2012, Iowa State University.

46 Garg, N., Raman spectroscopy for characterizing and determining the pozzolanic reactivity of fly ashes.

47 Gunasekaran, S., et al., Raman and infrared spectra of carbonates of calcite structure.

48 Sharma, S., et al. Remote Raman spectroscopy of various MIXED and composite mineral phases at 7.2 m distance. in 37th Lunar and Planetary Science Conference. 2006.

49 Pérez-Alonso, M., et al., Analysis of bulk and inorganic degradation products of stones, mortars and wall paintings by portable Raman microprobe spectroscopy. Analytical and bioanalytical chemistry, 2004. 379: p. 42-50.

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No.	Composition	Sample A	Sample C	Ref.
3	Hydromagnesite $Mg_5(CO_3)_4(OH)_2 \cdot 4H_2O$	755	---	50
4	Iron oxide	240-380-480	240	52 51
5	Gypsum-anhydrite $CaSO_4 \cdot 2H_2O - CaSO_4$	410-1115-1150	420-490-615- 1010	54 53
6	SiO ₂ Quartz	790	805-1185	57 56 55
7	Whewellite-weddellite $CaC_2O_4 \cdot H_2O - CaC_2O_4 \cdot 2H_2O$	595-870-1485-1535	1465-1475	59 58
8	Na ₂ SO ₄ · 10H ₂ O Mirabilite	1195	---	60
9	CaNO ₃ · 4H ₂ O Calcium nitrate	1055-1425	1425	62 61
10	Carbon black	1360	1365	63
11	Anorthite $CaAl_2Si_2O_8$	---	975	64

50 Gázquez, F., et al., In situ Raman characterization of minerals and degradation processes in a variety of cultural and geological heritage sites. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, 2017. 172: p. 48-57.

51 Pérez-Alonso, M., et al., Analysis of bulk and inorganic degradation products of stones, mortars and wall paintings by portable Raman microprobe spectroscopy.

52 Mendonça Filho, F., et al., In-situ versus laboratory characterization of historical site in marine environment using X-ray fluorescence and Raman spectroscopy. *Microchemical Journal*, 2019. 147: p. 905-913.

53 Garg, N., Raman spectroscopy for characterizing and determining the pozzolanic reactivity of fly ashes.

54 Maguregui, M., et al., Field Raman analysis to diagnose the conservation state of excavated walls and wall paintings in the archaeological site of Pompeii (Italy). *Journal of Raman Spectroscopy*, 2012. 43(11): p. 1747-1753.

55 Sharma, S., et al. Remote Raman spectroscopy of various MIXED and composite mineral phases at 7.2 m distance.

56 Palanivel, R. and G. Velraj, FTIR and FT-Raman spectroscopic studies of fired clay artifacts recently excavated in Tamilnadu, India. 2007.

57 Pérez-Alonso, M., et al., Analysis of bulk and inorganic degradation products of stones, mortars and wall paintings by portable Raman microprobe spectroscopy.

58 Palanivel, R. and G. Velraj, FTIR and FT-Raman spectroscopic studies of fired clay artifacts recently excavated in Tamilnadu, India.

59 Pérez-Alonso, M., et al., Analysis of bulk and inorganic degradation products of stones, mortars and wall paintings by portable Raman microprobe spectroscopy.

60 Maguregui, M., et al., Field Raman analysis to diagnose the conservation state of excavated walls and wall paintings in the archaeological site of Pompeii (Italy).

61 Pérez-Alonso, M., et al., Analysis of bulk and inorganic degradation products of stones, mortars and wall paintings by portable Raman microprobe spectroscopy.

62 Maguregui, M., et al., Field Raman analysis to diagnose the conservation state of excavated walls and wall paintings in the archaeological site of Pompeii (Italy).

63 Maguregui, M., et al., Field Raman analysis to diagnose the conservation state of excavated walls and wall paintings in the archaeological site of Pompeii (Italy).

64 Sharma, S.K., B. Simons, and H. Yoder, Raman study of anorthite, calcium Tschermak's pyroxene, and gehlenite in crystalline and glassy states. *American Mineralogist*, 1983. 68(11-12): p. 1113-1125.

No.	Composition	Sample A	Sample C	Ref.
12	Phosphates	1255	1250	65
13	Ettringite $\text{Ca}_6\text{Al}_2(\text{SO}_4)_3(\text{OH})_{12}\cdot 26\text{H}_2\text{O}$	---	555	66
14	Birnessite $\text{MnO}_2\cdot n\text{H}_2\text{O}$	---	570	67
15	Calcium phosphate	---	1060	68
16	Halite NaCl	305-345	340	70 69

3.1.4.2 Mortar

On the other hand, the analysis results for sample B (the mortar) indicate that it consists of hydraulic lime and sand. The presence of calcite, dolomite, belite, alite, gypsum, and calcium silicate hydrate compounds, which are the primary components of hydraulic lime, was noted⁷¹. Additionally, halite was found as one of the weathering agents (Table 6) (Figure 10). The presence of calcium acetates suggests the weathering of calcium carbonates by organic acids (acetic acid). Known anthropogenic sources of atmospheric organic acids include direct emission from motor vehicles, acetic acid reacts with calcite to form calcium acetate^{72 73 74}.

65 Kuwik, M., J. Pisarska, and W.A. Pisarski, Influence of oxide glass modifiers on the structural and spectroscopic properties of phosphate glasses for visible and near-infrared photonic applications. *Materials*, 2020. 13(21): p. 4746.

66 Garg, N., Raman spectroscopy for characterizing and determining the pozzolanic reactivity of fly ashes.

67 Gázquez, F., et al., In situ Raman characterization of minerals and degradation processes in a variety of cultural and geological heritage sites.

68 Ostrooumov, M., A Raman, IR and XRD analysis of the deterioration on historical monuments: Case study from Mexico. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, 2009. 73(3): p. 498-504.

69 Parro, V., et al., A microbial oasis in the hypersaline Atacama subsurface discovered by a life detector chip: implications for the search for life on Mars. *Astrobiology*, 2011. 11(10): p. 969-996.

70 Ostrooumov, M., A Raman, IR and XRD analysis of the deterioration on historical monuments: Case study from Mexico.

71 Callebaut, K., et al., Nineteenth century hydraulic restoration mortars in the Saint Michael's Church (Leuven, Belgium): Natural hydraulic lime or cement? *Cement and Concrete Research*.

72 Prince, Amy Preszler, et al. "Reactive uptake of acetic acid on calcite and nitric acid reacted calcite aerosol in an environmental reaction chamber." *Physical Chemistry Chemical Physics* 10.1 (2008): 142-152.

73 Franke, Walter A. "The durability of rocks—developing a test of rock resistance to chemical weathering." *American Journal of Science* 309.8 (2009): 711-730.

74 Ryu, Hwa-Sung, et al. "Steel-corrosion characteristics of an environmental inhibitor using limestone sludge and acetic acid." *International Journal of Concrete Structures and Materials* 12 (2018): 1-9.

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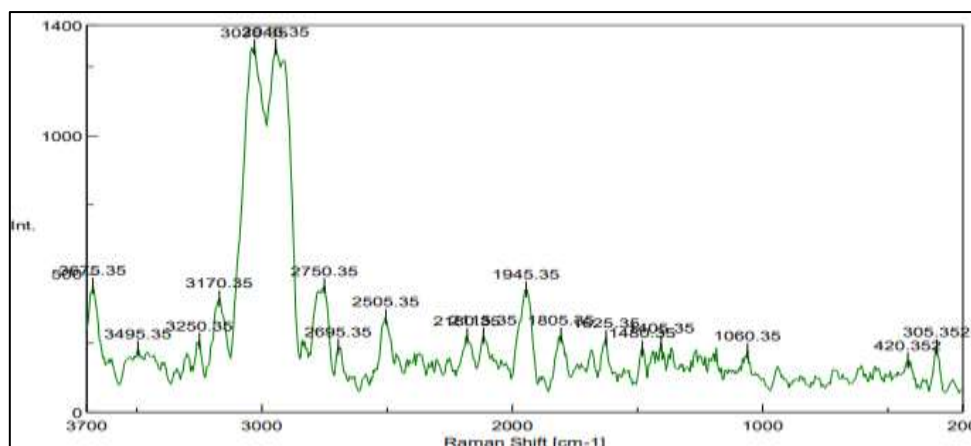


Figure 10 Results of Raman Spectroscopy: for sample B (binding mortar)

Table 6 shows the Raman results for the construction mortar (sample B) and includes the chemical composition, characteristic wavelengths, and main references.

No.	Composition	Sample B	Ref.
1	CaCO ₃ Calcite	1080-1090	75
2	CaMg(CO ₃) ₂ Dolomite	305-735-1760	76
3	Gypsum-anhydrite CaSO ₄ .2H ₂ O-CaSO ₄	420-3495	77
4	SiO ₂ Quartz	460	
5	C ₂ S Belite	510-520-535-805-865-1105-1120-1375	
6	C ₃ S Alite	835-845-850-880-1340-1365	
7	CaAl ₂ SiO ₇ Gehlenite	550-660	78
8	CSH Calcium silicate hydrate	360-660-680	79
9	Carboxylic Acid	1625	80
10	Magnesium acetate tetrahydrate	1070-2945-1425	81

75 Garg, N., Raman spectroscopy for characterizing and determining the pozzolanic reactivity of fly ashes.

76 Gunasekaran, S., et al., Raman and infrared spectra of carbonates of calcite structure.

77 Garg, N., Raman spectroscopy for characterizing and determining the pozzolanic reactivity of fly ashes.

78 Sharma, S.K., et al., Raman study of anorthite, calcium Tschermak's pyroxene, and gehlenite in crystalline and glassy states.

79 Garg, N., Raman spectroscopy for characterizing and determining the pozzolanic reactivity of fly ashes.

80 Hadjiivanov, K.I., et al., Power of infrared and raman spectroscopies to characterize metal-organic frameworks and investigate their interaction with guest molecules. Chemical Reviews, 2020. 121(3): p. 1286-1424.

81 Cohen, S., et al., A new double salt: triacetatohexaaquodimagnesium nitrate. Journal of crystal growth, 2003. 254(1-2): p. 151-155.

11	Calcium acetate	940-1480-2930-3030-3525	83 82
12	Calcium phosphate	1060	84
13	NaCl Halite	345-310	

3.2 Examination and Analysis of Saint Eugénie Church Building Materials

3.2.1 Imaging and Examination with Digital USB Microscopy

The Figures captured with the USB Digital Microscope (USB D.M) for the stone sample (D) show the cohesion of sand grains (quartz) bonded together by calcareous deposits in an integrated structure, indicating that they are sandy limestones. The surface irregularities and loss of surface parts due to weathering and decomposition are also observed^{85 86}. The brick sample (E) shows fine grains, with some sand grains and internal cracks and voids that may result from deterioration and damage.

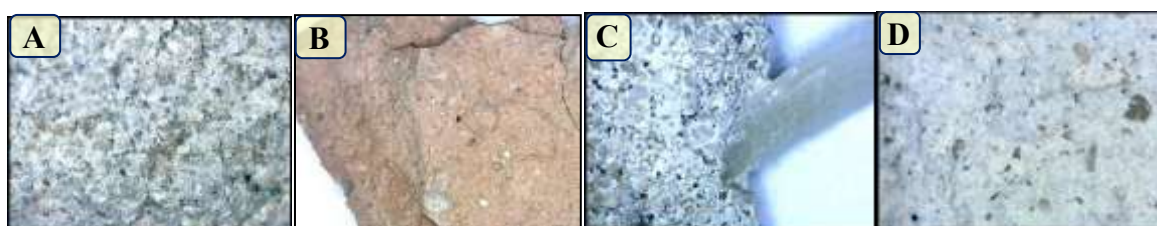


Figure 11: USB D.M: (a) Sample D, building stone, (b) Sample E, brick, (c) Sample F, bonding mortar, (d) Sample G, exterior wall plaster.

The bonding mortar (F) consists of a binding material that could be cement or hydraulic lime due to its dark gray color, with sand as a filler material. The presence of marine shell remains and shells in the mortar indicates that the sand used in construction might be sea sand, possibly sourced from the city's beachfront. The internal structure of the plaster (G) shows the binding material with sand grains (Figure 11a, b, c, d).

3.2.2 SEM (EDX) Analysis

The results of the EDX analysis, according to Table 7, show that oxygen and calcium (Ca, O) are the primary elements in sample D, with silicon as a secondary component represented by quartz grains. The presence of magnesium indicates

82 Pupo, Y.M., et al., Effect of a novel quaternary ammonium methacrylate polymer (QAMP) on adhesion and antibacterial properties of dental adhesives. *International journal of molecular sciences*, 2014. 15(5): p. 8998-9015.

83 Musumeci, A.W., et al., A spectroscopic study of the mineral paceite (calcium acetate). *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, 2007. 67(3-4): p. 649-661.

84 Ostrooumov, M., A Raman, IR and XRD analysis of the deterioration on historical monuments: Case study from Mexico.

85 Adefris, D., et al., Petrography and stable oxygen and carbon isotopic composition of the Antalo Limestone, Mekelle Basin, Northern Ethiopia: implications for marine environment and deep-burial diagenesis.

86 Fronteau, G., et al., Limestone millstones: Facies, provenance and use of sandy to pure limestones in France. *Journal of Lithic Studies*, 2020. 7(3): p. 20 p.-20 p.

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dolomite ($\text{CaMg}(\text{CO}_3)_2$) but in a lower proportion than calcite, confirming that the stone used is sandy limestone. Sodium and chlorine indicate the presence of halite salt.

The main components of sample E, representing the construction brick, are calcium, likely due to high calcium carbonate content in the clay or the use of calcium carbonate during manufacturing to improve brick properties⁸⁷. The clay minerals consist of silicon with aluminum, potassium, iron, and magnesium. The high concentration of sodium and chlorine in the sample results from significant halite salt deposition on the surface.

For sample F, the primary components were oxygen, silicon, and calcium, with aluminum, iron, magnesium, and potassium as secondary elements, indicating that the mortar used is not ordinary aerial lime. These elements are typical of the clay added during mortar manufacturing to impart hydraulic properties⁸⁸. The external plaster sample (G) from the western facade mainly consists of oxygen, calcium, and sulfur, indicating the use of gypsum and lime. Gypsum crystals are clearly visible (Figure 13e), and the presence of silicon suggests quartz (sand) was used in the plaster (Figure 12a, b, c, d).

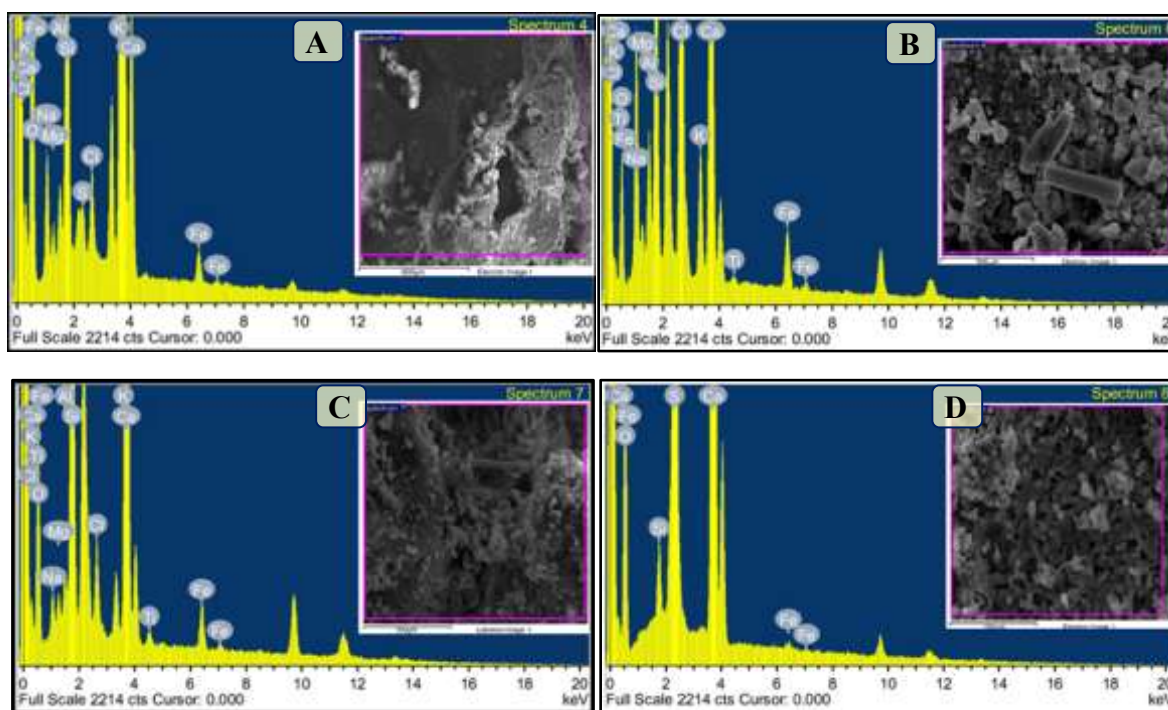


Figure 12: SEM-EDX Analysis Results: (a) for sample D (stone), (b) for sample E (brick), (c) for sample F (mortar), (d) for sample G (plaster).

⁸⁷Martirena, J., et al., Improvement of engineering properties of fired clay bricks through the addition of calcite, 7th IMC. 2006, London.

⁸⁸Figueiredo, C., M. Lawrence, and R. Ball. Chemical and physical characterisation of three NHL 2 binders and the relationship with the mortar properties.

Table 7 shows the EDX results for samples D, E, F, and G:

Sample	O (%)	Ca (%)	Si (%)	Mg (%)	Fe (%)	Al (%)	Cl (%)	Na (%)	K (%)	S (%)	Ti (%)
D	60.9	22.4	6.1	1.1	0.75	1.3	1	4.6	1.7	0.25	-
E	36.9	14.3	10.2	2	2.5	3.8	14.5	12.7	2.9	-	0.27
F	56.3	15.8	16.4	1.3	2	1.35	2.4	2.7	1.3	-	0.43
G	63.6	22.4	1.2	-	0.2	-	-	-	-	12.6	-

The examination and imaging using SEM for sample D (Figure 13a) reveal the fine structure of the stone's fabric. We see the quartz grains, indicated by number 1, bonded by calcareous deposits of calcium carbonate (CaCO_3), indicated by number 2. One of the skeletal remains of an organism is indicated by number 3, and the arrow points to the deposition of halite crystals on the surface. The distribution of pores and cracks in the stone's internal structure is also observed, which may be due to decomposition and weathering processes.

The fine structure of sample E shows homogeneity and precision (Figure 13b-c), indicating that the brick production process was at suitable temperatures of (800-950) degrees Celsius⁸⁹. Another Figure of the sample shows the dense deposition of halite crystals (NaCl) on the surface, indicated by number 1 (Figure 13c), suggesting that the main cause of brick deterioration and damage is the high salt content, along with the availability of moisture sources from groundwater, rain, and water vapor.

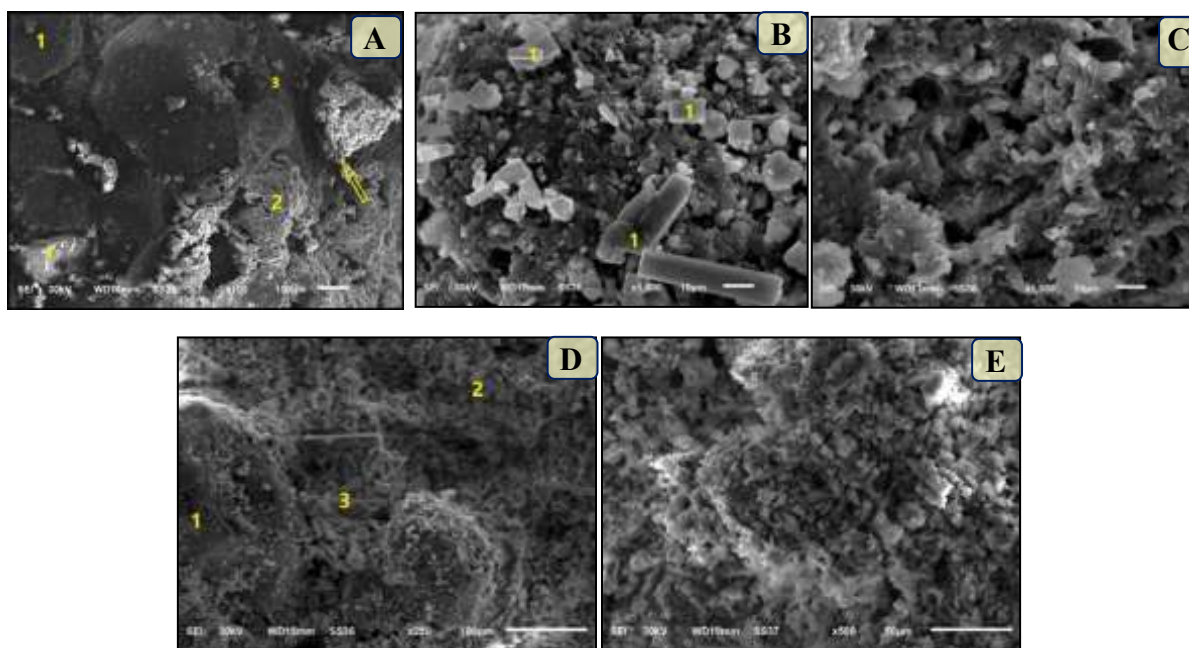


Figure 13: SEM Examination: (a) Sample D, building stone, where number 1 indicates quartz, 2 indicates calcite deposits, 3 indicates an organic remain, and the arrow points to halite,

⁸⁹ Saenz, N., et al., Analysis of tempered bricks: from raw material and additives to fired bricks for use in construction and heritage conservation. *European Journal of Mineralogy*, 2019. 31(2): p. 301-312.

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(b) Sample E, brick, showing pores in the fine structure resulting from weathering, (c) Sample E, brick, showing halite salt deposits on the brick surface, indicated by number 1, (d) Sample F, bonding mortar, where number 1 indicates quartz, 2 indicates calcium silicate, and 3 indicates halite, (e) Sample G, exterior wall plaster

Sample F mortar consists of sand (quartz) represented by number 1 in Figure 13d, bonded together by the binding material (calcium silicate) represented by number 2. Halite crystals (NaCl) are also deposited on the surface, indicated by number 3. Gypsum crystals can be seen clearly in sample G, representing the exterior plaster of the western facade, as shown in Figure 13e.

3.2.3 X-Ray Fluorescence (XRF) Analysis

Elemental analysis of the building stone samples reveals a high percentage of calcium oxide (CaO 59.8% $\pm$ 0.09%), while silicon oxide (SiO_2) indicates the stone contains sand (15.3% $\pm$ 0.26%). Magnesium oxides suggest the presence of dolomite in the stone ($\text{CaMg}(\text{CO}_3)_2$), confirming that the stone used in construction is sandy limestone⁹⁰. The chlorine percentage in the sample indicates the presence of halite salt (NaCl). The concentrations of other metal oxides like iron, aluminum, and potassium were low, indicating they are impurities within the stone. The presence of sulfur is attributed to weathering processes, as mentioned earlier.

The analysis results for sample E (brick) show a high percentage of silicon oxides (31.8% $\pm$ 0.34%), with silicates as the main component of the clay minerals used in brick manufacturing. Calcium oxides (22.3% $\pm$ 0.11%) suggest that the clay used was rich in calcium carbonate or that calcium carbonate was used in the brick manufacturing process to improve its properties. Other metal oxides appeared as secondary compounds within the clay minerals. The high concentration of chlorine (35.4% $\pm$ 0.05%) in the brick sample indicates significant halite salt (NaCl) deposition.

The analysis results for sample F, the mortar used in constructing the church, show that silicon oxide (50.2% $\pm$ 0.48%) and calcium oxide (40.1% $\pm$ 0.09%) are the main components. The high silicon oxide percentage indicates the use of sand in the mortar and the presence of silicates as compounds within the clay minerals used in mortar manufacturing. Aluminum, magnesium, iron, and potassium oxides were present as secondary compounds, confirming that the mortar used was not ordinary aerial lime. The C.I. index for the mortar used in the church's construction indicates it is cement, with a value of 3.345. The chlorine percentage in the sample indicates the presence of halite salt (NaCl).

The analysis results for sample G, representing the external plaster of the western wall, show that calcium oxides and sulfur are the main components (48.7% $\pm$ 0.09% and 33.5% $\pm$ 0.1%, respectively), indicating the use of lime and gypsum. Silicon oxide appeared as a

90 Fronteau, G., et al., Limestone millstones: Facies, provenance and use of sandy to pure limestones in France.

secondary compound ($11.2\% \pm 0.28\%$), meaning the plaster mortar was made of lime, gypsum, and sand, used in a ratio of approximately 3:1 binding material to sand, based on the component percentages (Figure 14a, b, c, d) table 8.

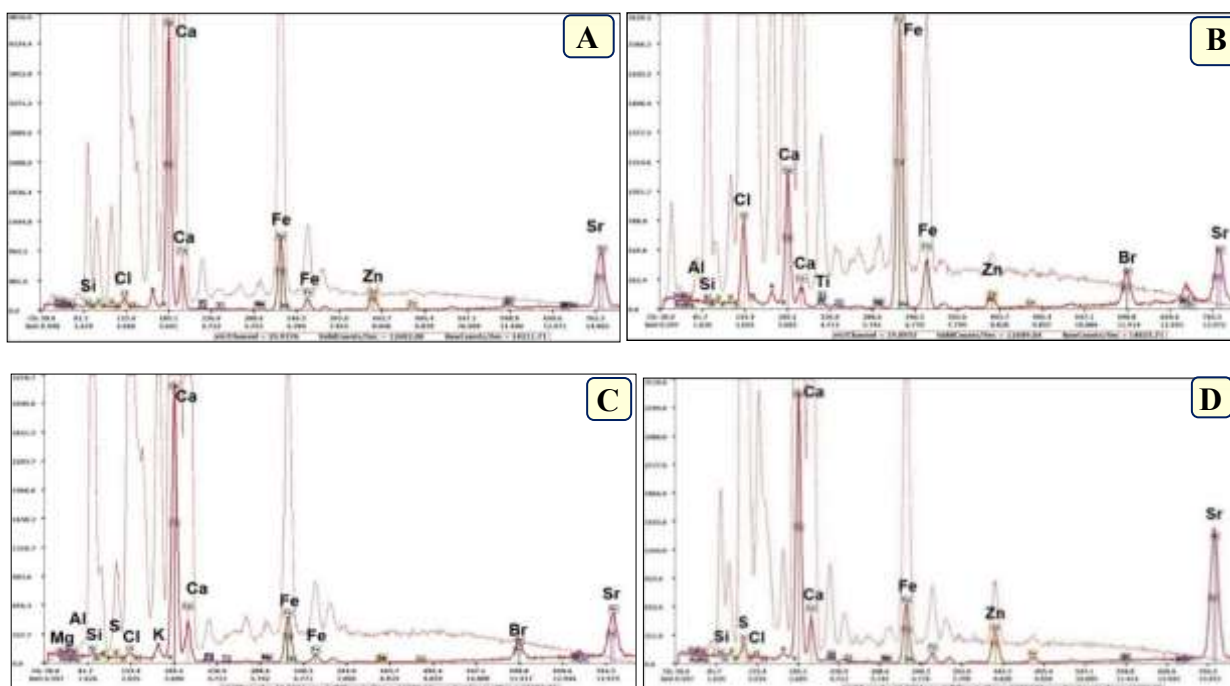


Figure 14: XRF Pattern Results: (a) Sample D (stone), (b) Sample E (brick), (c) Sample F (mortar), (d) Sample G (plaster).

Table 8 shows the XRF analysis results for samples D, E, F, and G:

Comp.	SiO ₂	CaO	Al ₂ O ₃	MgO	Fe	P ₂ O ₅	S	K ₂ O	Cl	others
D	15.3	59.8	3.04	4.99	1.89	1.63	1.3	3.8	7.38	0.87
E	31.8	22.3	4.97	0	3.06	0.23	0	1.39	35.4	0.85
F	50.2	40.1	1.5	1.78	0.71	0.26	0.58	1.7	2.96	0.21
G	11.2	48.7	2.5	0	1.39	0.79	33.5	0.28	0.51	1.13

3.2.4 Raman Spectroscopy Analysis

3.2.4.1 Stones and Fired Brick

The Raman spectroscopy analysis results for sample D, the building stone, indicate that the stones used in construction are sandy limestone, containing sand grains. The minerals calcite, dolomite, quartz, and calcium phosphate are present, along with iron oxides as impurities. The results for sample E, the brick, show it is composed of calcite, dolomite, iron oxides, gehlenite, and anorthite. Halite salt is observed in both samples as a weathering agent (Table 9) (Figure 15a, b).

It is worth noting that the presence of gypsum, calcium oxalate, calcium nitrate, and calcium acetates indicates the chemical weathering of the stone^{91 92}. The presence of

91 Allen, G., et al., Role of NO₂ and SO₂ in degradation of limestone.

92 Cezar, T.M., Calcium oxalate: a surface treatment for limestone.

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calcite and dolomite phases in the brick suggests their use in the manufacturing process to improve the brick's properties. The presence of gehlenite and anorthite in the brick indicates that the firing temperature during manufacturing was approximately 800-950 degrees Celsius⁹³ as mentioned earlier.

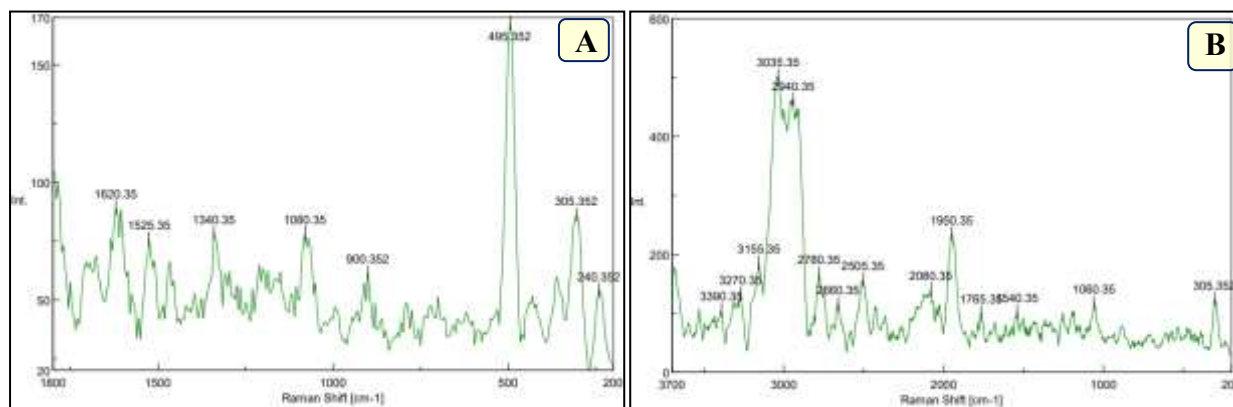


Figure 15: Raman Spectroscopy Results: (a) Sample D (stone), (b) Sample E (brick).

Table 9 shows the Raman analysis results for the building stone and brick (samples D-E), including the chemical composition, characteristic wavelengths, and key references.

No.	Composition	Sample D	Sample E	Ref.
1	Calcite-Aragonite-Vaterite CaCO_3	700-1080-1085- 1090-1635	1080-1085- 1090	94 95
2	Dolomite $\text{CaMg}(\text{CO}_3)_2$	305-735-1105- 1455-1770	305-1765	96 97
3	Iron oxide	240-295	240-295-410- 615-1540	98 99
4	Gypsum $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	490-495-1145	1010-1145	100 101

93 Saenz, N., et al., Analysis of tempered bricks: from raw material and additives to fired bricks for use in construction and heritage conservation.

94 Garg, N., Raman spectroscopy for characterizing and determining the pozzolanic reactivity of fly ashes.

95 Topuz, G., et al., Partial high-pressure aragonitization of micritic limestones in an accretionary complex, Tavşanlı Zone, NW Turkey. *Journal of Metamorphic Geology*, 2006. 24(7): p. 603-613.

96 Gunasekaran, S., et al., Raman and infrared spectra of carbonates of calcite structure.

97 Sharma, S., et al. Remote Raman spectroscopy of various MIXED and composite mineral phases at 7.2 m distance.

98 Klaes, B., et al., Iron (hydr) oxide formation in Andosols under extreme climate conditions. *Scientific Reports*, 2023. 13(1): p. 2818.

99 Mendonça Filho, F., et al., In-situ versus laboratory characterization of historical site in marine environment using X-ray fluorescence and Raman spectroscopy.

100 Garg, N., Raman spectroscopy for characterizing and determining the pozzolanic reactivity of fly ashes.

101 Sharma, S., et al. Remote Raman spectroscopy of various MIXED and composite mineral phases at 7.2 m distance.

No.	Composition	Sample D	Sample E	Ref.
5	Quartz SiO ₂	1340-1525	---	102
6	Whewellite CaC ₂ O ₄ .H ₂ O	1620	---	
7	Calcium nitrate Ca(NO ₃) ₂ .4H ₂ O	1050-1425	1425	103 104
8	Anorthite CaAl ₂ Si ₂ O ₈	---	570-1100	105
9	Gehlenite Ca ₂ Al ₂ SiO ₇	---	895	
10	Calcium phosphate	1060	1060	106
11	Halite NaCl	310-345	310	107 108
12	Acetylene	---	1950	109
13	Calcium acetate	---	1475-2930-3035	110
14	Magnesium acetate tetrahydrate	---	950-2940	111
15	Sulphuric acid	900	---	112
16	Magnesium nitrate hydrate Mg(NO ₃) ₂ .6H ₂ O	---	3390	113

3.2.4.2 Building Mortar and Wall Plaster

The analysis of the building mortar (sample F) reveals the presence of alite and calcium silicate hydrate phases in multiple peaks, with the disappearance of belite and

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- 102 Pérez-Alonso, M., et al., Analysis of bulk and inorganic degradation products of stones, mortars and wall paintings by portable Raman microprobe spectroscopy.
- 103 Maguregui, M., et al., Field Raman analysis to diagnose the conservation state of excavated walls and wall paintings in the archaeological site of Pompeii (Italy).
- 104 Pérez-Alonso, M., et al., Analysis of bulk and inorganic degradation products of stones, mortars and wall paintings by portable Raman microprobe spectroscopy.
- 105 Sharma, S.K., B. Simons, and H. Yoder, Raman study of anorthite, calcium Tschermak's pyroxene, and gehlenite in crystalline and glassy states.
- 106 Ostrooumov, M., A Raman, IR and XRD analysis of the deterioration on historical monuments: Case study from Mexico.
- 107 Parro, V., et al., A microbial oasis in the hypersaline Atacama subsurface discovered by a life detector chip: implications for the search for life on Mars.
- 108 Ostrooumov, M., A Raman, IR and XRD analysis of the deterioration on historical monuments: Case study from Mexico.
- 109 Czapinski, E.C., et al., Experimental characterization of the pyridine: acetylene co-crystal and implications for Titan's surface. *ACS Earth and Space Chemistry*, 2023. 7(3): p. 597-608.
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gehlenite phases, indicating the presence of cement¹¹⁴. Calcite, dolomite, and quartz are also present, suggesting that the mortar used is a mix of cement and sand. The external plaster sample (sample G) consists of calcite, gypsum, and sand, confirming that it is composed of lime, gypsum, and sand (see Figure 16a, b) (see Table 10).

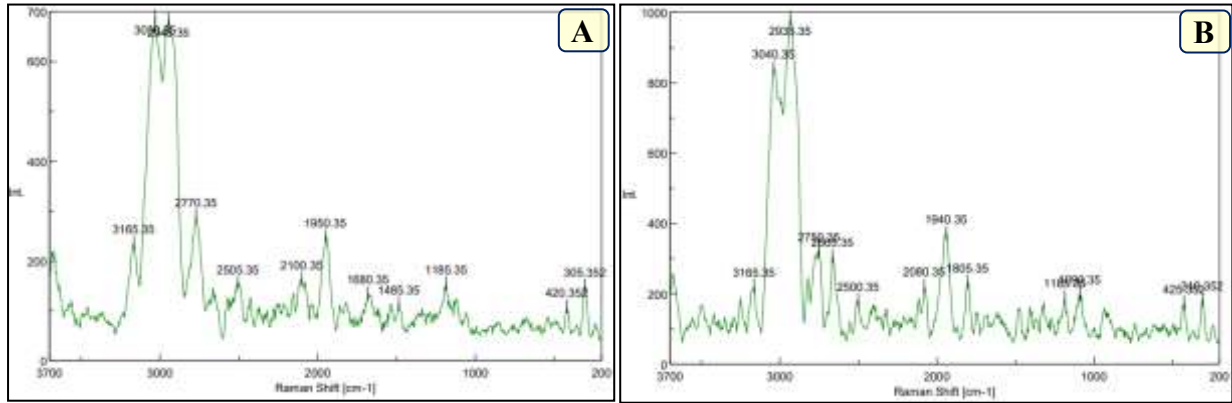


Figure 16: Raman Spectroscopy Results: (a) Sample F (mortar), (b) Sample G (plaster).

Table 10 shows the Raman analysis results for the mortar and plaster (samples F-G), including the chemical composition, characteristic wavelengths, and key references.

No.	Composition	Sample F	Sample G	Ref.
1	Calcite CaCO_3	1080	1090	115
2	Dolomite $\text{MgCa}(\text{CO}_3)_2$	305-1455-1765	310-735-1455	116
3	Gypsum-anhydrite $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}-\text{CaSO}_4$	420	425-435-1110-1125-1170-3495	117
4	Quartz SiO_2	1185	1185(Q) 1805(Si-OH)	118 119
5	$\text{Mg}_3(\text{HCOO})_6$	1680	---	120
6	Alite C_3S	845-850-855-1340-1360	---	121

114 Varas, M., M.A. De Buergo, and R. Fort, Natural cement as the precursor of Portland cement: Methodology for its identification. *Cement and Concrete Research*, 2005. 35(11): p. 2055-2065.

115 Garg, N., Raman spectroscopy for characterizing and determining the pozzolanic reactivity of fly ashes.

116 Gunasekaran, S., et al., Raman and infrared spectra of carbonates of calcite structure.

117 Garg, N., Raman spectroscopy for characterizing and determining the pozzolanic reactivity of fly ashes.

118 Pérez-Alonso, M., et al., Analysis of bulk and inorganic degradation products of stones, mortars and wall paintings by portable Raman microprobe spectroscopy.

119 Rkiouak, L., et al., Optical trapping and Raman spectroscopy of solid particles.

120 Hadjiivanov, K.I., et al., Power of infrared and raman spectroscopies to characterize metal-organic frameworks and investigate their interaction with guest molecules.

121 Garg, N., Raman spectroscopy for characterizing and determining the pozzolanic reactivity of fly ashes.

7	Calcium silicate hydrate CSH	355-490-525-620-680	---	
8	Whewellite $\text{CaC}_2\text{O}_4 \cdot \text{H}_2\text{O}$	1485	---	122
9	Magnesium acetate tetrahydrate	945-1060-2945	1425-2935	123
10	Calcium acetate	1405-1440-3030	965-1475-2935-3040	124 125
11	Acetylene	1950	1940	126
12	Halite NaCl	310	310-350	127 128

The presence of calcium oxalate and calcium acetate, indicates the chemical weathering of building materials, while the presence of halite suggests the salt-induced deterioration mechanism.

4. Discussion

Through this study, which used several techniques, starting with digital microscopy, scanning electron microscopy, X-ray fluorescence, and Raman spectroscopy, to determine the chemical composition and internal structure of the building materials used in both the de Lesseps base and the Saint Eugenie Church, it was concluded that limestone was the main building material in the base of the statue. Two types of limestone were used: (Micrite) limestone in building the base, and (Biosparites) limestone in building the promenade attached to the base, with the use of hydraulic lime mortar to bind the stone blocks. The examination revealed that the base building materials suffer from the risk of external factors of damage, such as wind, rainfall, humidity, temperature, and salts (halite), due to the coastal environment of Port Said. Air pollution gases and aerosols also played an important role in the chemical weathering of the stones and mortar, as the presence of calcium sulfate, calcium oxalate, and calcium nitrate indicates the decomposition of limestone components, which will lead to the weakening of the building materials.

On the other hand, we find that (Sandy limestone) was used side by side with fired bricks in the construction of the Church of Saint Eugenie. It appears through

122 Topuz, G., et al., Partial high-pressure aragonitization of micritic limestones in an accretionary complex, Tavşanlı Zone, NW Turkey.

123 Cohen, S., L. Ben-Dor, and Y. Marcus, A new double salt: triacetatohexaaquodimagnesium nitrate.

124 Pupo, Y.M., et al., Effect of a novel quaternary ammonium methacrylate polymer (QAMP) on adhesion and antibacterial properties of dental adhesives. International journal of molecular sciences, 2014. 15(5): p. 8998-9015.

125 Musumeci, A.W., R.L. Frost, and E.R. Wacławik, A spectroscopic study of the mineral paceite (calcium acetate).

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127 Parro, V., et al., A microbial oasis in the hypersaline Atacama subsurface discovered by a life detector chip: implications for the search for life on Mars.

128 Ostrooumov, M., A Raman, IR and XRD analysis of the deterioration on historical monuments: Case study from Mexico.

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examination that the bricks used were of a good quality, as it was found that they contained calcium carbonate, which was added to the bricks to improve their properties, in addition to the use of appropriate firing temperatures. A mortar of sand and cement was used, as that period witnessed the widespread use of cement in construction work. It was also found that the sand used was sea sand, as it contained the remains of sea shells, which proves that it was brought from the seashore, while the cladding of some of the external facades was composed of lime mortar, gypsum, and sand. The building materials of the church, like their counterparts at the base, suffer from the risk of external factors such as rain, wind, relative humidity, heat, salts (halite salt), and atmospheric pollution gases, which led to the weakness of the internal structure and the decomposition of the original components of the building materials.

From the above, it is clear that the continued exposure of building materials to the danger of external factors will inevitably lead to their weakness and damage, which necessitates the need to intervene to carry out the necessary maintenance work to preserve the survival and continuity of these important historical buildings.

5. Conclusion

This research involves an analytical study of the building materials used in constructing the base of the de Lesseps statue and the Saint Eugénie Church in Port Said. Examination and analysis concluded that the base of the de Lesseps statue was built using recrystallized micrite, calcite limestone. In contrast, the promenade (base walkway) was constructed with biosparites limestone. The bonding mortar was composed of hydraulic lime and sand, which are well-suited for the coastal environment.

Conversely, Saint Eugénie Church in Port Said was constructed using sandy limestone alongside fired brick. The brick quality was ensured by using appropriate firing temperatures during manufacture. The mortar used in the construction consisted of sand and cement, and the exterior walls were coated with plaster made of lime, sand, and gypsum. It is notable that sea sand was used in the mortar mix.

Salts, particularly halite, are the common factor in the deterioration of building materials. The cause is attributed to the coastal environment conditions, including high humidity rates, temperature fluctuations, and proximity to the sea. The impact of air pollution gases combined with moisture on the chemical weathering of building materials was also confirmed.

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